GRAVITY, MOTION, AND ORBITS

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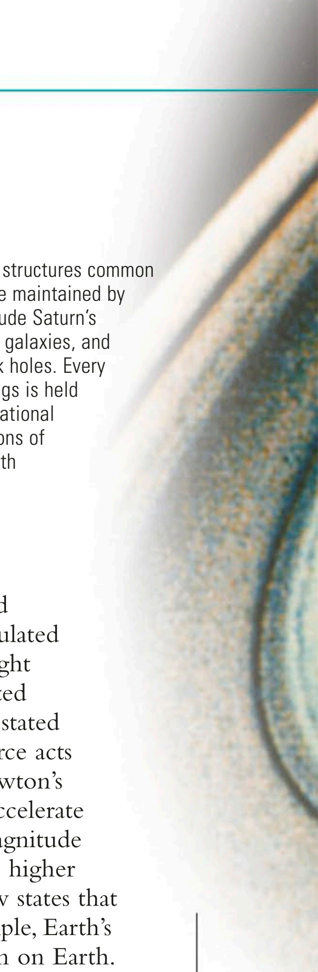
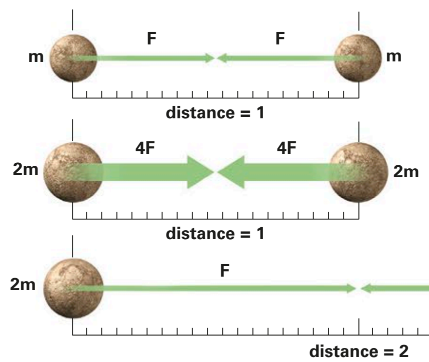
GRAVITY IS THE ATTRACTIVE FORCE that exists between every object in the universe, the force that both holds stars and galaxies together and causes a pin to drop. Gravity is weaker than nature's other fundamental forces, but because it acts over great distances and between all bodies possessing mass, it has played a major part in shaping the universe. Gravity is also crucial in determining orbits and creating phenomena such as planetary rings and black holes.



DISKS AND RINGS
The disk- and ringlike structures common in celestial objects are maintained by gravity. Examples include Saturn's rings (pictured), spiral galaxies, and the disks around black holes. Every particle in Saturn's rings is held in orbit through gravitational interactions with billions of other particles and with Saturn itself.

NEWTONIAN GRAVITY

The scientific study of gravity began with Galileo Galilei's demonstration, in about 1590, that objects of different weight fall to the ground at exactly the same, accelerating rate. In 1665 or 1666, Isaac Newton realized that whatever force causes objects to fall might extend into space and be responsible for holding the Moon in its orbit. By analyzing the motions of several heavenly objects, Newton formulated his law of universal gravitation. It stated that every body in the universe exerts an attractive force (gravity) on every other body and described how this force varies with the masses of the bodies and their separation. To this day, Newton's law remains applicable for understanding and predicting the movements of most astronomical objects.



1 Two bodies, each of mass m, attract each other with force F

2 Doubling the mass of each body while maintaining their separation quadruples the gravitational force to 4F

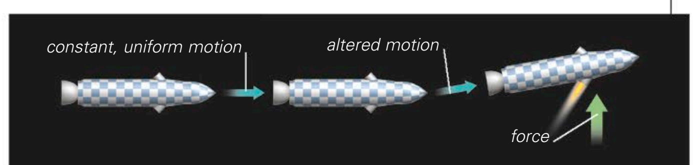
3 Doubling the separation between the bodies reduces the force by a factor of 4, back to F

MASS AND DISTANCE
Any two bodies are attracted by a force of gravity proportional to the mass of one multiplied by the mass of the other. The force is also inversely proportional to the square of their separation.

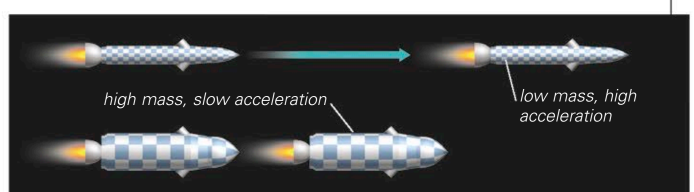
NEWTON'S LAWS OF MOTION

From his studies of gravity and the motions of heavenly bodies, and again extending concepts first developed by Galileo, Newton formulated his three laws of motion. Before Galileo and Newton, people thought that an object in motion could continue moving only if a force acted on it. In his first law of motion, Newton contradicted this idea: he stated that an object remains in uniform motion or at rest unless a net force acts on it. (A net force is the sum of all forces acting on an object.) Newton's second law states that a net force acting on an object causes it to accelerate (change its velocity) at a rate that is directly proportional to the magnitude of the force. It also states that the smaller the mass of an object, the higher the acceleration it experiences for a given force. Newton's third law states that for every action, there is an equal and opposite reaction—for example, Earth's gravitational pull on the Moon is matched by the pull of the Moon on Earth.

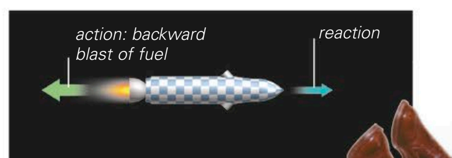
FIRST LAW OF MOTION
The first law states that an object remains in a state of rest or moves at a constant speed in a straight line unless acted on by a net force.



SECOND LAW OF MOTION
When an object of low mass and one of greater mass are subjected to a force of the same magnitude, the low-mass object accelerates at a higher rate.

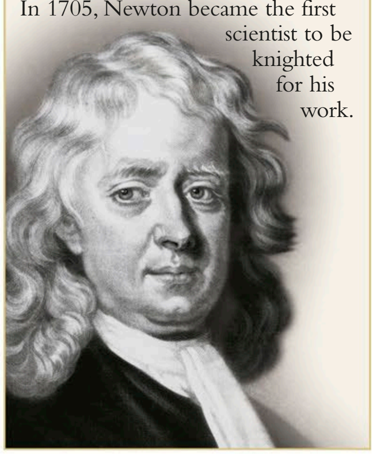


THIRD LAW OF MOTION
To every action, there is an equal and opposite reaction. The forward thrust of a rocket is a reaction to the backward blast of combusted fuel.



ISAAC NEWTON

The English mathematician and physicist Isaac Newton (1642–1727) was one of the greatest-ever scientific intellects. As well as his discoveries in the fields of gravity and motion, he co-discovered the mathematical technique of calculus. In 1705, Newton became the first scientist to be knighted for his work.



WEIGHT AND FREE FALL

The size of the gravitational force acting on an object is called its weight. An object's rest mass (measured in pounds or kilograms) is constant, while its weight (measured in newtons) varies according to the local strength of gravity. A mass of 2.2 lb (1 kg) weighs 9.8 newtons on Earth but only 1.65 newtons on the Moon. Weight can be measured, and the feeling of weight experienced, only when the gravity producing it is resisted by a second, opposing force. A person standing on Earth feels weight not so much from the pull of gravity as from the opposing push of the ground on his or her feet. In contrast, a person orbiting Earth is actually falling toward Earth under gravity. Such a person is in "free fall" and experiences apparent weightlessness. This is due not to lack of gravity but to the absence of a force opposing gravity.

WEIGHTLESSNESS
Astronauts in training must frequently experience apparent weightlessness. Here, a plane is plunging sharply from high altitude, putting the trainee astronauts inside into free fall.

INTRODUCTION